

How P2P Works

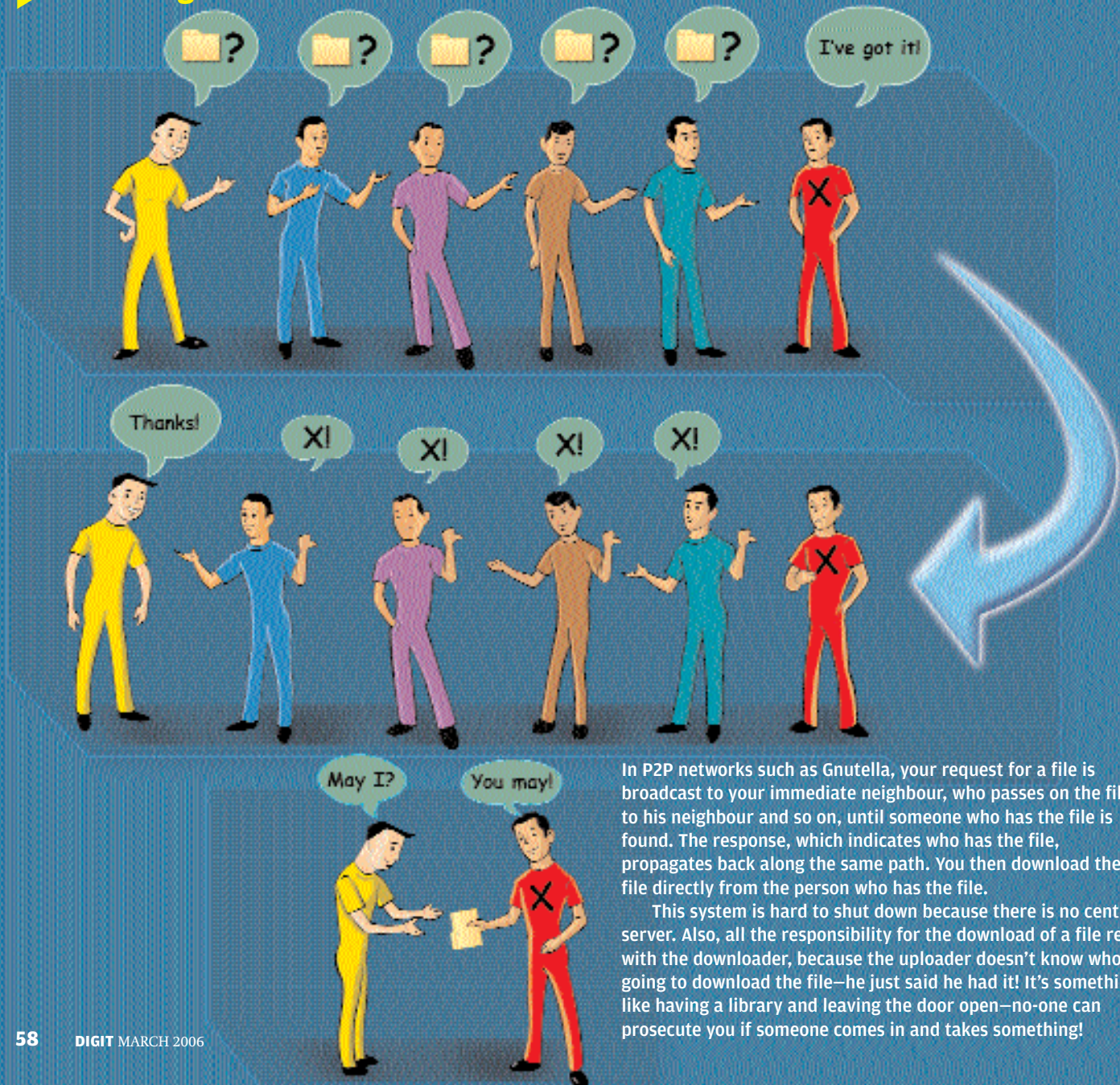
► How Napster Worked

In Napster, a centralised server had information about who had what files. You just asked the server, and it would point you to the person who had the file. And you then downloaded the file directly from that person.

The system was easy to shut down because of the centralised server: it actively helped people establish connections!



► How Regular P2P Works



In P2P networks such as Gnutella, your request for a file is broadcast to your immediate neighbour, who passes on the file to his neighbour and so on, until someone who has the file is found. The response, which indicates who has the file, propagates back along the same path. You then download the file directly from the person who has the file.

This system is hard to shut down because there is no central server. Also, all the responsibility for the download of a file rests with the downloader, because the uploader doesn't know who is going to download the file—he just said he had it! It's something like having a library and leaving the door open—no-one can prosecute you if someone comes in and takes something!